GRADE 2 · BLOCK CODING

Two Sprites at Once

One green flag starts two scripts

Name: .... Date: ....

★ **I can read two scripts that run at the same time.**

Two sprites, one green flag

Bit (green) and Pixel (purple) are two sprites. Each one has its own script. When you click the green flag ▶, BOTH scripts run at the SAME time. That is called concurrent — two things happening at once. Each sprite moves on the number path.

The number path

0	1	2	3	4	5
	Bit				Pixel

Bit

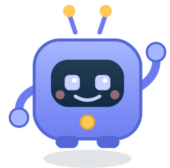
Bit's script

▶ start at 1

forward 2

1 Where does Bit land?

Run Bit's script from 1. What number does Bit land on?



Pixel

Pixel's script

▶ start at 5

back 1

2 Where does Pixel land?

Run Pixel's script from 5. What number does Pixel land on?

3 Predict — both at once

The green flag starts BOTH scripts together. Write where EACH sprite lands. Bit: ____ Pixel: ____. Do they land on the same number?

Big idea

Concurrent means two scripts run at the same time. Each sprite follows its own script, so each one lands on its own number when the green flag starts them together.

TEACHER GUIDE · GRADE 2 · BLOCK CODING

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Quick facilitation notes & answer key

What this worksheet teaches

Bit and Pixel are two characters, each with their own set of moves. One green flag starts BOTH at the same time. Students read each set of moves and find where each character lands. Two sets of moves running at once is the new Grade 2 idea. No coding experience needed.

Before you start (no computer needed)

No computer needed — paper and a pencil. You read the prompts aloud. Optional: put two counters on a number line 0 to 5 — green for Bit, purple for Pixel — and move BOTH at the same time so children see the two sets of moves run side by side.

How to lead it (about 20 minutes)

1. Show them (you do). Run Bit's moves alone, then Pixel's alone. Then say: "the green flag starts both at the same time."
2. Do one together. Exercise 1 — Bit lands on 3; Exercise 2 — Pixel lands on 4.
3. Let them try. Students write where each lands when both start, and whether they meet (Ex 3).

Answer key (these are the verified correct answers)

Exercise 1 — Bit: 1, forward 2 → 3. Exercise 2 — Pixel: 5, back 1 → 4.

Exercise 3 — the green flag starts both at once: Bit lands on 3, Pixel on 4. Different numbers, so they do not meet.

Key idea: the two sets of moves run on their own, at the same time — Bit's moves do not change Pixel's, and the green flag starts both together.

If students get stuck — and ways to adjust

Watch for: thinking one character's moves affect the other. Each one only moves itself.

Make it easier: move a green counter for Bit and a purple counter for Pixel, one move each.

Make it harder: ask how far apart Bit and Pixel are at the end.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

In plain terms: students write and run instructions, including two sets that run at the same time.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

In plain terms: students read two sets of instructions, change one, and explain what happens when both run together.

You'll know it worked when

The child finds where Bit and Pixel each land and tells whether they meet.